

Credible on Paper? Green Banking Policy, Access to Finance, and Institutional Capacity in the Greening of Firms Worldwide

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Abstract

Green banking policy frameworks coordinated through the Sustainable Banking and Finance Network (SBFN) have been adopted across more than sixty emerging and developing economies since 2012, on the premise that supervisory guidelines directing banks to price environmental risk into lending will strengthen the link between firms' access to credit and their green investment behavior. We test this premise using two independent World Bank Enterprise Survey samples. The primary sample harmonizes the Bank's full 2018–2020 Green Economy Module rollout across 41 economies (28,042 firms) on a rich seven-item green-practice-adoption outcome, matched to each economy's SBFN policy-adoption status and Worldwide Governance Indicators regulatory quality as of its survey year. Because SBFN status is fixed within country in a single-wave cross-section, a classical country-fixed-effects specification cannot identify its main effect, and identifies its interaction with credit access only off the eight adopter economies in that sample; we show this explicitly, then re-estimate the relationship with a Bayesian hierarchical model (country-varying slopes) and a causal forest that impose no linear-interaction functional form. All three approaches agree: bank credit access

is robustly associated with green practice adoption (average marginal effect of 13 percentage points in the baseline specification), but neither SBFN policy adoption nor regulatory quality moderates it—both institutional variables instead shift the general level of green adoption directly. We then subject this finding to the sharpest test we can construct: a second, independently assembled sample of 162 further economies (89,797–100,115 firms depending on specification), spanning 2021–2026 and reaching, for the first time in this literature, into high-income economies such as Australia, Germany, and Japan, built from a World Bank standardized cross-country database not previously exploited for this purpose. On a narrower CO₂-emissions-monitoring outcome and with country fixed effects the primary sample’s single-wave design cannot support informatively, credit access remains significant while both SBFN and regulatory-quality interactions remain statistically indistinguishable from zero—the identical qualitative pattern, on a different outcome, a different set of economies, and a materially sharper identification strategy. The causal forest’s feature-importance decomposition attributes the bulk of genuine effect heterogeneity in the primary sample (75.6%) to firm size rather than to either institutional moderator (regulatory quality 19.9%; SBFN status 2.6%). One finance-instrument-specific check finds a nominally significant negative credit $\times$ SBFN interaction when finance access is measured via overdraft rather than term credit, though this does not survive a Benjamini–Hochberg correction for multiple comparisons across the full family of interaction tests reported in the paper, and we treat it as suggestive rather than established. We interpret the combined evidence as more consistent with green banking policy’s binding constraint operating through

firm-level absorptive capacity than through the country-level regulatory capacity its own institutional design targets, though we report this as our best current account of an auxiliary, exploratory finding rather than as a settled result. Finally, because none of the above designs exploits exogenous variation in SBFN adoption *timing*, we additionally build a country-year panel (217 economies, 2000–2024) from World Bank World Development Indicators and estimate a Callaway-Sant’Anna staggered event-study around each economy’s own adoption date – the identification strategy this literature most often calls for. It shows no effect on renewable-energy share and a positive association with CO2 emissions per capita that shrinks by roughly 45% once GDP per capita is controlled for and whose event-study profile is more consistent with pre-existing differences in adopting economies’ growth trajectories than with a causal effect of adoption; this independent, differently-identified test converges on the same substantive conclusion as the firm-level analysis.

Keywords: green banking policy, access to finance, institutional complementarity, firm-level green investment, causal forest, hierarchical Bayesian model

JEL: G21, O16, Q56, D22, C11

1. Introduction

Since 2012, more than sixty central banks, banking regulators, and financial-sector regulatory bodies across the emerging and transition world have signed on to a coordinated effort to green their financial systems, adopting national guidelines that direct commercial banks to price environmental and climate risk into lending decisions and to channel credit toward cleaner pro-

7 duction. On paper, this represents one of the most ambitious diffusions of
8 environmental financial regulation outside the OECD: economies as differ-
9 ent in income, banking-sector depth, and political regime as Bangladesh,
10 Morocco, Mongolia, and Indonesia now report, to the International Finance
11 Corporation’s Sustainable Banking and Finance Network (SBFN), formal
12 sustainable-finance roadmaps, green-taxonomy pilots, or mandatory environmental-
13 risk disclosure rules for lenders. Whether that adoption changes what actu-
14 ally happens on a factory floor—whether a firm that can get a bank loan is
15 any more likely to install a wastewater treatment system, switch to a more
16 efficient boiler, or measure its own emissions—is a separate and considerably
17 harder question, and it is the one this paper answers.

18 The premise that formal adoption of a policy is not the same as its op-
19 eration in practice is, of course, not new to institutional economics. A large
20 literature on state capacity and policy credibility has long argued that the
21 same law, regulation, or guideline can produce entirely different real-economy
22 outcomes depending on the administrative and judicial infrastructure avail-
23 able to enforce it (La Porta et al., 1998; Acemoglu et al., 2001). Applied to
24 green finance specifically, a parallel body of work has documented that green
25 credit guidelines in China measurably reshape lending allocation and firm
26 environmental performance (Li et al., 2023; Huang et al., 2023; Dai et al.,
27 2025), that firms’ green investment responds to the financing constraints they
28 face (Ghosh and Dutta, 2022; Costa et al., 2024; De Haas et al., 2025), and
29 that the World Bank Enterprise Surveys’ own Green Economy module is a
30 workable lens on firm-level environmental behavior across dozens of countries
31 at once (Asif et al., 2025; Phan, 2026). What is missing is the connection

32 between these two literatures: a direct test, at firm level and across a wide
33 cross-section of the very economies the green-banking-policy movement tar-
34 gets, of whether the credit-to-green-investment channel that these guidelines
35 are designed to activate is actually stronger where the guidelines exist—and
36 whether that depends on the state capacity to make the guideline credible in
37 the first place.

38 This paper closes that gap using firm-level data from two independently
39 assembled samples of World Bank Enterprise Survey (WBES) waves. The
40 primary sample covers 47 country-years fielded between 2018 and 2023 that
41 carry the Bank’s full Green Economy module—a battery of questions on emis-
42 sions monitoring, energy audits, and the adoption of climate-friendly produc-
43 tion measures—alongside the Surveys’ standard access-to-finance module; a
44 second, global sample extends this to 162 further country-years fielded be-
45 tween 2021 and 2026, reaching for the first time in this literature into high-
46 income economies with no SBFN involvement at all (Section 4). We match
47 each economy to its Sustainable Banking and Finance Network policy status
48 as of its survey year, distinguishing genuine ex-ante policy adopters from
49 economies that joined only afterward, and to the World Bank’s Worldwide
50 Governance Indicators as a measure of the regulatory state capacity avail-
51 able to make that policy credible. Econometrically, we depart deliberately
52 from the pooled fixed-effects specification that has become the default in
53 this literature. Because SBFN status is fixed within country in a single-
54 wave cross-section, a country fixed effect absorbs its main effect entirely, and
55 would identify its interaction with credit access—which remains estimable,
56 being a firm-level term—only for the eight adopter economies in that sample;

57 a naive interaction term estimated by pooled logit, which we report first as
 58 a baseline precisely to make this limitation visible rather than to hide it,
 59 returns no detectable policy-moderation effect at all. We show that this null
 60 result is not evidence of an absent relationship so much as a symptom of
 61 forcing a fundamentally hierarchical question (does a level-2, country-level
 62 policy variable reshape a level-1, firm-level financing relationship?) into a
 63 flat, single-level model with too few country clusters to lean on for inference
 64 (Cameron and Miller, 2015). We then re-estimate the same relationship with
 65 tools built for exactly this structure: a Bayesian hierarchical logistic model
 66 with country-varying slopes (Gelman and Hill, 2007; Gelman et al., 2020), a
 67 causal-forest / double-machine-learning (DML) estimator (Wager and Athey,
 68 2018; Chernozhukov et al., 2018; Athey et al., 2019; Athey and Imbens, 2019)
 69 that lets the treatment effect of credit access vary flexibly across the institu-
 70 tional landscape rather than through a single estimated coefficient, and, on
 71 the much larger global sample, a country-fixed-effects specification that the
 72 primary sample’s single wave cannot support with useful precision.

73 This paper contributes to the literature in six respects. First, it pro-
 74 vides what is, to our knowledge, the first direct empirical test of whether
 75 SBFN-style green banking policy adoption changes the firm-level relation-
 76 ship between finance and green investment, rather than only the aggregate
 77 volume of green lending reported by banks themselves—a firm-level, demand-
 78 side complement to the bank-level, supply-side evidence that dominates the
 79 existing green-credit-policy literature. Second, it extends the World Bank
 80 Enterprise Survey’s Green Economy module—used until now mostly within
 81 single countries or single regional waves—into a harmonized 41-country pri-

82 mary sample spanning Europe, Central Asia, the Caucasus, and the Middle
 83 East and North Africa, giving the paper a breadth of institutional varia-
 84 tion that a single-country or single-region design cannot offer. Third, and
 85 unusually for this literature, we subject the primary sample’s central find-
 86 ing to an independently assembled second test: a 162-economy global sam-
 87 ple built from a World Bank standardized cross-country database spanning
 88 2021–2026 that, to our knowledge, no published study of green banking pol-
 89 icy has yet exploited, extending coverage for the first time into high-income
 90 economies entirely outside the SBFN’s own remit and allowing a country-
 91 fixed-effects specification the primary sample’s single-wave design cannot
 92 support. That the same qualitative result—a robust finance-green channel,
 93 no detectable institutional moderation—survives this independent replica-
 94 tion is, we think, considerably more informative than either sample could
 95 be alone. Fourth, methodologically, the paper demonstrates—with the same
 96 underlying question tested twice, on two different samples, under four dif-
 97 ferent estimators—exactly where and why a classical fixed-effects approach
 98 fails to detect (or, when country fixed effects are usable, correctly fails to
 99 find) an institutionally-conditioned effect, offering applied researchers work-
 100 ing with similarly structured cross-country firm data a concrete illustration
 101 of when the extra modeling investment is worth making and when a null
 102 result is simply a null result. Fifth, the paper redirects the question of which
 103 system-level margin governs green-credit-policy effectiveness: having ruled
 104 out a country-institutional interaction across four estimators and two sam-
 105 ples, we show (Section 3.4) that the effect heterogeneity that does exist is
 106 concentrated at the level of firm size rather than country institutions, relo-

107 cating the operative institutional margin from the country-regulatory level to
108 the firm-scale level—a mechanism we introduce as an auxiliary, exploratory
109 hypothesis in Section 3.4 and demonstrate empirically in Section 5.4 rather
110 than treat as a pre-registered finding, and are correspondingly explicit about
111 the identification limits (Section 4) that a purely cross-sectional design, how-
112 ever extensively replicated, still carries. Sixth, and directly in response to
113 that identification-strategy limit, Section 6 builds an entirely independent
114 country-year panel from World Bank World Development Indicators and es-
115 timates a Callaway-Sant’Anna staggered event-study exploiting SBFN’s own
116 adoption-timing variation—the design class this literature most often asks
117 for—and finds that it, too, does not support a beneficial institutional ef-
118 fect on aggregate environmental outcomes, converging with the firm-level
119 evidence from a design built specifically to meet an identification bar the
120 firm-level data cannot.

121 The paper also speaks directly to a growing body of recent scholarship on
122 green finance, credit access, and institutional quality in emerging and tran-
123 sition economies. Jia et al. (2026) study green credit policy and sustainable
124 production across the BRICS economies and find a positive long-run relation-
125 ship that is itself strengthened where governance is stronger—a country-level
126 result whose firm-level analogue is precisely what our H2 and H3 test. Chi
127 and Yang (2023) exploit China’s 2016 green-financial-system guidelines in a
128 difference-in-differences design and find a green transition effect concentrated
129 where state environmental governance is strong, and Shi et al. (2024), across
130 37 Asian economies, find that green finance improves environmental outcomes
131 only where control of corruption is high. W. Ullah et al. (2024) establish the

132 more general moderation logic we adopt, showing that regulatory quality con-
133 ditions the financial-development-to-growth relationship rather than merely
134 shifting its level, and Jin (2026) documents, across thirty economies at firm
135 level, that firms may decline formal credit altogether—a demand-side mar-
136 gin our realized-access measure cannot observe. Read together, this literature
137 has established that green-finance effectiveness is institutionally conditional
138 at the country level and that credit access is institutionally conditioned at
139 the firm level. What has not been tested, and what this paper contributes,
140 is whether the two combine as the theory implies: whether a dateable green-
141 banking-policy adoption event moderates the finance-to-green-investment re-
142 lationship at firm level, across a broad cross-section of adopting economies,
143 and whether that moderation depends on regulatory capacity.

144 The remainder of the paper is organized as follows. Section 2 sets out
145 the institutional background of the SBFN framework and the concepts and
146 trends in green banking policy adoption across our sample. Section 3 devel-
147 ops the paper’s three hypotheses from the state-capacity and institutional-
148 complementarity literature. Section 4 describes the primary and global sam-
149 ples, variable construction, and the empirical strategy for each. Section 5
150 presents baseline, hierarchical, and heterogeneous-effect results on the pri-
151 mary sample, followed by the independent global-sample replication. Sec-
152 tion 7 discusses the findings’ economic magnitude and their implications for
153 how green credit policy is designed and sequenced, and Section 8 concludes.

154 **2. Institutional background: green banking policy in emerging and** 155 **transition economies**

156 The Sustainable Banking and Finance Network (SBFN)—established in
157 2012 as the Sustainable Banking Network and renamed in 2019 to reflect an
158 expanded mandate covering both banking and non-bank financial regulation—
159 is the principal vehicle through which financial-sector regulators and banking
160 associations in emerging and developing economies have coordinated the de-
161 sign of national sustainable-finance policy. Hosted and technically supported
162 by the International Finance Corporation, the network now counts roughly
163 seventy member institutions (central banks, banking regulators, and banking
164 or securities associations) drawn overwhelmingly from Sub-Saharan Africa,
165 East Asia and the Pacific, Europe and Central Asia, Latin America and
166 the Caribbean, the Middle East and North Africa, and South Asia (Inter-
167 national Finance Corporation / Sustainable Banking and Finance Network,
168 2025). Notably for our sample, the network’s membership is essentially dis-
169 joint from the European Union: not one of the sixteen EU member states in
170 our WBES sample appears on any SBFN membership roster we could locate,
171 consistent with the network’s explicit mandate to serve markets outside the
172 reach of the EU’s own sustainable-finance architecture (the Sustainable Fi-
173 nance Disclosure Regulation and the EU Taxonomy, both of which post-date
174 most of our survey waves in any case).

175 Membership itself, however, is only the network’s coarsest measure of
176 policy adoption. SBFN’s own Multi-Tiered Progress Framework—the in-
177 strument used in its periodic Global Progress Reports (2018, 2019, 2021,
178 2023–2024) and country-level Progress Report addenda—scores each mem-

179 ber jurisdiction’s actual regulatory output against several policy pillars, most
 180 centrally environmental-and-social risk management and, in later report vin-
 181 tages, ESG disclosure integration and green-finance-flow measurement, and
 182 places each jurisdiction on a progression from an initial “Commitment” or
 183 “Formulating” stage, through “Developing” or “Implementation,” to “Advanc-
 184 ing” and, for the most mature frameworks, “Consolidating.” The variation
 185 this produces across our sample is considerable and is itself part of the paper’s
 186 identifying variation, not just a robustness afterthought. Mongolia’s Bank
 187 of Mongolia was a founding SBFN member in 2012 and had already reached
 188 the network’s “Advancing” implementation stage by the time of the country’s
 189 2019 Enterprise Survey; Morocco’s Bank Al-Maghrib, a member since 2014,
 190 sat at the same stage two years later. Bangladesh Bank, also a founding 2012
 191 member, had progressed by its 2022 Enterprise Survey wave into the net-
 192 work’s ESG-integration pillar at an “Advancing” classification, while India’s
 193 Reserve Bank/Indian Banks’ Association, a later (2016) entrant, remained
 194 at a “Developing” commitment-pillar stage at its own 2022 survey. Jordan
 195 and the Kyrgyz Republic, both members since 2016 and 2018 respectively,
 196 had progressed only as far as an initial “Commitment” stage by their 2019
 197 survey waves. A further, and for identification purposes equally important,
 198 group of economies in our sample joined SBFN only after their Enterprise
 199 Survey was fielded—Kazakhstan (2021), Serbia and Ukraine (2020), Tajik-
 200 istan (2023), and most of the Western Balkan and South Caucasus economies
 201 in our sample (2021–2024)—and are accordingly coded as non-adopters as of
 202 their survey year, even though several are now members. This ex-ante cod-
 203 ing is deliberate: it is the policy environment a firm actually operated under

204 at the moment its Green Economy module responses were recorded that is
205 relevant to our test, not the policy environment that jurisdiction eventually
206 adopted.

207 The economic logic behind the SBFN framework, and the reason it is a
208 plausible channel for firm-level green investment rather than a purely sym-
209 bolic commitment, rests on a straightforward supervisory mechanism: once
210 a banking regulator formally adopts environmental-and-social risk manage-
211 ment guidelines, commercial banks under its supervision face a compliance
212 incentive to price environmental risk into individual lending decisions and,
213 in many of the more advanced frameworks, to report a green-lending share
214 to the supervisor directly. Existing evidence on China’s much-studied green
215 credit guidelines documents exactly this channel operating at bank and firm
216 level—reduced lending to high-polluting borrowers and improved environ-
217 mental performance among firms that retain access to credit (Li et al., 2023;
218 Huang et al., 2023; Dai et al., 2025; Gao et al., 2022)—and a smaller liter-
219 ature on Bangladesh’s own long-running green-banking guidelines points in
220 the same direction (Khairunnessa et al., 2021). What neither literature has
221 established is whether this channel is detectable across a broad, heteroge-
222 neous cross-section of SBFN-style adopters at the firm level using a common
223 survey instrument, nor whether the strength of the channel depends on the
224 regulatory capacity available to enforce the underlying guideline—the two
225 questions this paper turns to next.

226 **3. Literature review and hypothesis development**

227 *3.1. Access to finance and firm-level green investment*

228 A firm’s decision to install pollution-abatement equipment, retrofit ma-
229 chinery for energy efficiency, or absorb the upfront cost of an emissions-
230 monitoring system is, first and foremost, an investment decision, and in-
231 vestment decisions are shaped by whether a firm can finance them. The
232 theoretical case is close to a standard credit-constraint story: green invest-
233 ments of the kind captured in the WBES Green Economy module (heat-
234 ing and cooling upgrades, on-site renewable generation, waste-minimization
235 retrofits, energy-management systems) typically carry a large upfront capi-
236 tal outlay against a payback stream realized only over several years, which is
237 exactly the profile of investment most sensitive to a firm’s access to external
238 finance rather than to retained earnings alone. Firm-level evidence bears
239 this out directly. Ghosh and Dutta (2022), using a panel of Indian manu-
240 facturing firms, show that credit-constrained firms are measurably less likely
241 to adopt cleaner production practices even when they report an equivalent
242 willingness to comply with environmental regulation. De Haas et al. (2025)
243 document, across a large cross-country firm sample, that managerial and fi-
244 nancial barriers operate as distinct but jointly binding constraints on firms’
245 green transition investments, and Kalantzis et al. (2022) reach a closely re-
246 lated conclusion using EBRD transition-economy survey data, finding that
247 firms’ green investment is driven by a mix of financing conditions and cli-
248 mate exposure rather than by climate motivation alone—precisely the joint
249 finance-and-institutions framing this paper tests directly. Costa et al. (2024)
250 and Truong (2026) find, respectively in OECD and Vietnamese data, that

251 financing constraints blunt firms’ green investment response to environmen-
 252 tal policy pressure that would otherwise induce it, while Aristei and Gallo
 253 (2023) show that firms with better access to credit are both more likely to
 254 have adopted green management practices before a shock and better able to
 255 sustain them through one. B. Ullah (2025) narrows this further to unlisted
 256 SMEs specifically, showing that financial constraints bind corporate environ-
 257 mental responsibility at smaller firm scales—a firm-size channel this paper’s
 258 own causal-forest evidence (Section 5) also implicates, though, as we report
 259 there, without recovering a clean monotonic size gradient of its own. Siewers
 260 et al. (2024) extend the firm-level environmental-performance literature to
 261 a global-value-chain setting, showing that a firm’s position in international
 262 production networks shapes its environmental performance net of domes-
 263 tic financing conditions, and Bambe et al. (2026) show that environmental
 264 policy stringency itself reshapes firm efficiency in developing economies, un-
 265 derscoring that the firm-level margin, not only the aggregate or bank-level
 266 one, is where much of this literature’s true test now lies. Taken together, this
 267 literature converges on a simple first-order proposition, which any test of a
 268 green-credit-policy channel must first establish before asking whether policy
 269 moderates it:

270 **H1.** Firms with access to formal bank credit (a credit line or
 271 loan from a financial institution) are more likely to have adopted
 272 green production practices than otherwise-similar firms without
 273 such access.

274 *3.2. Does green banking policy strengthen the finance-to-green-investment*
275 *channel?*

276 H1 describes a channel, not a policy. The entire premise of the SBFN
277 framework and its national analogues—and of the much larger literature on
278 green credit guidelines in China specifically (Li et al., 2023, 2024; Huang et
279 al., 2023)—is that a supervisory mandate directing banks to price environ-
280 mental risk into lending, or to preferentially finance green activity, should
281 widen the gap in green-investment behavior between firms with and without
282 bank access relative to a counterfactual world without the policy: banks in
283 adopting jurisdictions have a compliance incentive to seek out, and price
284 favorably, exactly the borrowers undertaking the kind of investment our
285 outcome variable measures, while credit-constrained firms in the same ju-
286 risdiction see no comparable improvement in their financing terms. This
287 is a moderation hypothesis, not a simple main-effect one, and it is worth
288 being explicit that most of the existing supporting evidence—again, over-
289 whelmingly from China (Dai et al., 2025; Gao et al., 2022) plus a smaller
290 literature on Bangladesh (Khairunnessa et al., 2021), Vietnam (Nguyen and
291 Phan, 2025), and cross-country fintech-credit channels (Tran et al., 2026)—
292 is drawn from single-country studies of a single, unusually well-resourced
293 green-credit regime. Whether the same interaction holds across the far more
294 heterogeneous set of jurisdictions that have adopted an SBFN-style frame-
295 work, many of them at a far earlier and less resourced stage of implementation
296 than China’s, is untested. This motivates:

297 **H2.** The positive association between bank credit access and
298 green-practice adoption (H1) is stronger in economies that had,

299 as of the survey year, adopted an SBFN sustainable-finance policy
300 framework than in economies that had not.

301 *3.3. Institutional complementarity: does policy credibility depend on state*
302 *capacity?*

303 A green banking guideline is only as effective as the supervisory apparatus
304 behind it. This is the institutional-complementarity argument at the center of
305 the comparative-institutions, law-and-finance, and state-capacity literatures.
306 Hall and Soskice (2001)’s varieties-of-capitalism framework is the founda-
307 tional statement of the underlying logic in its most general form: formally
308 similar policy instruments produce systematically different outcomes depend-
309 ing on how they interlock with the surrounding institutional architecture—
310 a regulatory mandate that functions as intended within one configuration
311 of complementary institutions may function quite differently, or not at all,
312 transplanted into another. Applied more narrowly to financial regulation
313 specifically, the same logic runs through the law-and-finance and state-capacity
314 literatures: La Porta et al. (1998) and the subsequent finance-and-growth
315 literature (Levine, 2005; Demirgüç-Kunt and Maksimovic, 1998; Beck et al.,
316 2005, 2008) established that formally identical financial-contracting rules pro-
317 duce very different real-economy financing outcomes depending on the quality
318 of the legal and regulatory institutions that enforce them, while Acemoglu
319 et al. (2001) showed the same logic operating at the level of a country’s en-
320 tire institutional endowment. Applied to our setting, the argument is direct:
321 a bank supervisor’s environmental-risk guideline can only reshape lending
322 behavior to the extent that the supervisor has the regulatory capacity—
323 examination authority, credible sanctions, reliable disclosure enforcement—

324 to make compliance with it more than nominal. Where that capacity is
 325 weak, an SBFN framework may exist on paper (a formal “Commitment” or
 326 “Formulating” stage classification, in the network’s own terminology) with-
 327 out meaningfully altering bank lending behavior on the ground, precisely the
 328 state-capacity/policy-credibility distinction this paper is built to test. The
 329 evidence that green finance works only through capable institutions is by
 330 now reasonably direct at the country level. Shi et al. (2024), across 37 Asian
 331 economies, find that green finance improves environmental outcomes where
 332 control of corruption is strong and has little detectable effect where it is weak;
 333 Chi and Yang (2023) find China’s green-financial-system guidelines drive
 334 firms’ green transition most where state environmental governance is strong;
 335 and Jia et al. (2026) find the long-run green-credit-to-sustainable-production
 336 relationship across the BRICS economies is itself strengthened by governance
 337 quality. Each of these is an aggregate or single-jurisdiction result, and each
 338 implies a firm-level prediction—that the credit-to-green channel should re-
 339 spond to policy more sharply under stronger regulatory capacity—that has
 340 not been tested directly on firm-level data across adopting economies. That
 341 is the prediction H3 states and this paper tests. We proxy regulatory ca-
 342 pacity with the World Bank’s Worldwide Governance Indicators Regulatory
 343 Quality estimate, matched to each economy’s survey year, and hypothesize
 344 a three-way interaction:

345 **H3.** The policy-moderation effect described in H2 is itself in-
 346 creasing in regulatory quality: SBFN adoption strengthens the
 347 credit-to-green-investment channel most where the supervising
 348 jurisdiction has the institutional capacity to make the underly-

349 ing guideline credible, and least—potentially not at all—where it
350 does not.

351 Hypotheses H2 and H3 are, by construction, statements about how a
352 country-level variable reshapes a firm-level relationship. That two-level struc-
353 ture is what Section 4 argues a classical single-level specification is poorly
354 placed to adjudicate on this sample—not because the interaction terms it
355 tests are unidentified, but because a country-cluster count in the low forties
356 makes cluster-robust inference on them unreliable, and because a country-
357 fixed-effects variant would purchase within-country identification at the cost
358 of the level-2 main effects and of resting on only eight adopter economies.
359 That is what motivates the hierarchical and machine-learning estimators we
360 turn to after reporting the classical benchmark.

361 *3.4. An auxiliary hypothesis: firm-level absorptive capacity as an alternative*
362 *margin*

363 H2 and H3 are both, in the end, statements that a *country-level* char-
364 acteristic conditions the strength of the credit-to-green channel. A separate
365 strand of the financing-constraints literature suggests the conditioning vari-
366 able that matters most may instead sit at the *firm* level: B. Ullah (2025)
367 shows that financial constraints bind corporate environmental responsibility
368 specifically at smaller firm scales, consistent with a standard fixed-cost logic
369 in which the up-front capital outlay a green retrofit requires (Section 3) is
370 proportionately heavier, and therefore harder to absorb even once financed,
371 for a smaller firm than a larger one. We did not set out to test this as a fourth
372 ex-ante hypothesis on a par with H1–H3; it emerged from the causal forest’s

373 feature-importance decomposition (Section 5.4), which we report honestly
374 here as an auxiliary, exploratory hypothesis motivated by that same fixed-
375 cost logic rather than one we pre-specified:

376 **H4 (auxiliary, exploratory).** Whatever heterogeneity exists
377 in the strength of the credit-to-green-adoption channel is more
378 attributable to firm size than to either country-level institutional
379 moderator (SBFN status or regulatory quality).

380 We flag its exploratory status explicitly because it changes how its evi-
381 dence should be read: Section 5.4’s support for H4 is genuine and, we think,
382 informative, but it is evidence discovered during analysis rather than con-
383 firmed against a pre-registered prediction, and the same section reports hon-
384 estly that it does not yet extend to a clean, monotonic size gradient. H4
385 also connects this paper’s evidence to the just-transition and SME-financing
386 literature’s broader concern with who is left behind when a policy instru-
387 ment operates on an extensive banking margin rather than on the firm-scale
388 margin that determines whether a financed firm can absorb a green invest-
389 ment’s fixed costs at all—a distributional dimension of green-credit-policy
390 design this paper’s data can only gesture toward, not test directly, but which
391 we think is a natural extension for future work using firm-level data with
392 a continuous size measure and a larger, more granular firm-size distribution
393 than the Enterprise Surveys’ own three-category strata provide.

394 4. Data and methodology

395 4.1. Data sources and sample construction

396 The paper’s firm-level data come from the World Bank Enterprise Sur-
397 veys (WBES), a standardized, stratified-random-sample survey of formal-
398 sector, non-agricultural, non-extractive private firms with at least five em-
399 ployees, conducted by the World Bank’s Enterprise Analysis Unit. Our pri-
400 mary analysis sample is the World Bank’s own harmonized cross-country
401 file combining the 41 Europe and Central Asia (ECA) and Middle East and
402 North Africa (MENA) economies surveyed under a single, identical question-
403 naire between 2018 and 2020, the survey wave in which the Bank’s Green
404 Economy module—a battery of questions on emissions monitoring, energy-
405 consumption targets, and the adoption of specific climate-friendly production
406 measures—was fielded in full to every respondent. This yields 28,042 firm-
407 level observations across 41 economies (Table 2). We supplement this primary
408 sample with 15,556 additional firm observations from six further economies
409 (Bangladesh 2022, India 2022, Indonesia 2023, Peru 2023, Philippines 2023,
410 Timor-Leste 2021) in which a shortened, partially randomized version of the
411 Green Economy module was fielded; because the exact item wording and ran-
412 domized sub-module assignment differ across these six economies, we do not
413 pool them into the main harmonized outcome variable, and instead use them
414 as an out-of-region external-validity check on a narrower, honestly-matched
415 pair of indicators (CO2-emissions monitoring, available in five of the six;
416 waste-minimization adoption, available in five of the six, with different spe-
417 cific economies covered by each), reported separately in Section 5.

418 The primary sample, while broad by the standards of the existing green-

419 credit-policy literature, is not the full universe of WBES economies carrying
 420 *any* Green Economy content, and we build a second, independent sample
 421 to exploit the rest of it. The World Bank also maintains a standardized
 422 cross-country database (2006–2026) that harmonizes a common set of core
 423 variables across every Formal Sector Enterprise Survey wave it has ever
 424 fielded, 292,314 firm-level observations across 471 country-years. Within
 425 it, 160 country-years fielded between 2021 and 2026 — reaching, notably,
 426 into high-income economies entirely outside the SBFN’s remit such as Aus-
 427 tralia, Canada, Germany, and Japan — carry a harmonized minimal Green
 428 Economy item set: climate-damage exposure and, more relevantly for our
 429 outcome construct, CO2-emissions monitoring, alongside the same standard
 430 access-to-finance module (credit line/loan, overdraft, perceived finance ob-
 431 stacle) used in the primary sample. We supplement this with Bangladesh
 432 2022 and Indonesia 2023 — two economies whose Green Economy responses
 433 used an alternative randomized-module variable naming convention and so
 434 are absent from the standardized database’s harmonized columns, but which
 435 we can recover from our own extraction of the underlying country files —
 436 for a combined global sample of 162 economies and 100,115 firms (89,797
 437 with complete data on all covariates). Firm size, sector, and productivity
 438 controls for this sample are drawn from the Bank’s companion Productivity
 439 Database, merged by firm identifier. We refer to this as the *global sample*,
 440 and to CO2-emissions monitoring as its outcome variable throughout; it is
 441 the paper’s vehicle for testing whether the primary sample’s central find-
 442 ing replicates independently, at far greater scale, and under an identification
 443 strategy — country fixed effects — that the primary sample’s single-wave

444 design cannot support (Section 4).

445 Country-level data are drawn from two sources for both samples. For the
446 primary sample’s 47 economy-years, SBFN policy-adoption status, join year,
447 and progression-framework stage as of each economy’s survey year are com-
448 piled from the Sustainable Banking and Finance Network’s Global Progress
449 Reports (2018, 2019, 2021, 2023–2024), country-level Progress Report ad-
450 denda, and the network’s live country-profile data portal, cross-checked against
451 IFC press releases announcing individual member accessions; economies that
452 joined SBFN only after their WBES survey year are coded as non-adopters
453 as of that year (Section 2). For the global sample’s remaining 157 economy-
454 years, we draw membership status and join year directly from the SBFN data
455 portal’s full published member roster (67 members as of the portal’s most
456 recent update), applying the identical ex-ante rule: an economy is coded as
457 an adopter only if its join year precedes its survey year. Six economy-years
458 in the global sample (Central African Republic, Cameroon, Chad, Republic
459 of Congo, Equatorial Guinea, and Gabon, all 2023–2024) are recorded on
460 the roster under a regional entry, “Central African States,” which we inter-
461 pret as referring to the CEMAC zone’s common banking supervisor; four
462 small-island economy-years (Antigua and Barbuda, Grenada, St. Lucia, St.
463 Vincent and the Grenadines, all 2025) are similarly recorded under “East-
464 ern Caribbean States,” which we interpret as the Eastern Caribbean Central
465 Bank’s member states. Both interpretations are flagged explicitly here as
466 inferences rather than explicit per-country listings. Regulatory capacity is
467 measured with the World Bank’s Worldwide Governance Indicators (WGI)
468 Regulatory Quality estimate, matched to each economy’s exact survey year

469 (or, where not yet published, the most recent prior year available) via the
470 World Bank DataBank API; WGI does not cover Taiwan, which is accord-
471 ingly dropped from global-sample regressions.

472 A natural alternative to the cross-sectional design used throughout this
473 paper would exploit SBFN’s own staggered adoption timing—join years in
474 our sample span 2012–2024 (Section 2)—in a staggered difference-in-differences
475 or event-study design around each economy’s own adoption date, the identi-
476 fication strategy this literature most often asks for. We do not attempt this,
477 and it is worth being explicit about why rather than leaving the gap implicit.
478 The WBES Green Economy module is not a repeated panel instrument: it
479 has been fielded to a given country, in most cases, exactly once, as a special
480 one-off module attached to a single survey wave, and even the small subset of
481 economies in our sample surveyed more than once by the WBES core instru-
482 ment were not re-surveyed with a comparable Green Economy module in a
483 second wave; the six supplementary economies in our own extension sample
484 (Section 5.5) illustrate the problem directly, since even they differ from one
485 another, and from the primary sample, in exact item wording and randomized
486 sub-module assignment, precluding a consistently defined outcome measured
487 twice for the same country before and after its SBFN join date. A within-
488 country before/after comparison around each economy’s own adoption date
489 is accordingly not implementable with a consistent outcome variable in the
490 currently published WBES microdata, for this specific outcome construct.
491 This is a genuine data-availability constraint on the identification strategy
492 available to us, not a design choice; we flag it here as a first-class limitation
493 of the paper’s identification, and return to its consequences in Section 8.

494 4.2. Variable construction

495 **Outcome.** Our primary dependent variable, *green practice adoption*, is
496 a binary indicator equal to one if a firm reports having adopted at least one
497 of seven Green Economy module practices over the preceding three years:
498 more climate-friendly on-site energy generation, an energy-management sys-
499 tem, waste-minimization or recycling measures, air-pollution control mea-
500 sures, other pollution-control measures, any energy-efficiency measure, or
501 the use of on-site renewable energy. We also construct a continuous *green*
502 *adoption index*, the mean of the same seven binary items, for robustness.

503 **Treatment.** *Bank credit access* is a binary indicator equal to one if the
504 firm reports holding a line of credit or loan from a formal financial institution
505 at the time of the survey.

506 **Country-level moderators.** *SBFN member* is a binary indicator equal
507 to one if the economy had adopted a national SBFN sustainable-finance
508 policy framework as of the survey year. *Regulatory quality* is the WGI Reg-
509 ulatory Quality estimate for the survey year (continuous, standardized to
510 mean zero, unit standard deviation within the estimation sample for the
511 hierarchical and machine-learning specifications).

512 **Controls.** Firm size (small/medium/large, per the Enterprise Surveys’
513 own sampling strata), broad sector (manufacturing, services, construction),
514 log sales, an exporter dummy (any direct or indirect export share), and a
515 foreign-ownership dummy (any private foreign ownership share).

516 Table 1 lists every variable used in the analysis together with its exact
517 WBES question code and source, in the interest of full transparency about
518 variable construction given the multi-source nature of the merge.

519 *4.3. Empirical strategy*

520 We estimate the relationship between bank credit access and green prac-
 521 tice adoption, and its moderation by SBFN status and regulatory quality,
 522 in five stages of increasing methodological sophistication, each addressing a
 523 specific limitation of the one before it.

524 **Stage 1 — classical benchmark.** We first estimate a pooled logistic
 525 regression,

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\left(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + \delta_{s(i)}\right) \quad (1)$$

526 for firm i in country c , where $\mathbf{X}_{i,c}$ is the firm-control vector and $\delta_{s(i)}$ is a sec-
 527 tor fixed effect, with standard errors clustered by country. We deliberately
 528 do not include country fixed effects in this specification, and it is worth being
 529 precise about what such a fixed effect would and would not absorb. Because
 530 SBFN_c is constant within country in a single cross-sectional wave, a coun-
 531 try fixed effect absorbs its main effect entirely, making β_2 unidentifiable by
 532 construction. The interaction $\text{Credit}_{i,c} \times \text{SBFN}_c$, by contrast, is a firm-level
 533 regressor and remains formally identified under country fixed effects, since
 534 credit access itself varies within country; Stage 5 below relies on exactly that
 535 logic to identify the corresponding interactions on the global sample. What
 536 the primary sample cannot support is not the identification of β_3 but its pre-
 537 cision: with country fixed effects, β_3 would be identified only by the contrast
 538 between within-country credit variation in the eight SBFN-adopter economies
 539 of this sample and that in the remaining thirty-three (Section 5), while the
 540 SBFN main-effect level shift those economies exhibit—a first-order feature
 541 of the data—would be absorbed and unreportable. We therefore estimate

542 β_2 and β_3 jointly without country fixed effects, as the classical benchmark
 543 against which Stage 2’s hierarchical alternative is compared, extended with
 544 a triple interaction for Regulatory quality_c that tests H3’s three-way claim
 545 directly.

546 **Stage 2 — Bayesian hierarchical (multilevel) model.** We re-
 547 estimate the same relationship allowing country to enter as a random rather
 548 than fixed effect, with a random slope on Credit:

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \quad (2)$$

$$\begin{aligned} \text{logit}(p_{i,c}) = & (\beta_1 + u_{1,c}) \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) \\ & + \beta_4 (\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + (\alpha + u_{0,c}) \end{aligned} \quad (3)$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (4)$$

549 estimated via Hamiltonian Monte Carlo (the No-U-Turn Sampler, NUTS)
 550 in PyMC, an open-source Python library for Bayesian statistical modeling
 551 (Salvatier et al., 2016). This specification is the paper’s primary vehicle for
 552 testing H2: because country enters through a variance component rather than
 553 a full set of dummies, the country-level moderators SBFN_c and RegQuality_c
 554 retain identified main effects, while the random slope $u_{1,c}$ still absorbs unob-
 555 served country heterogeneity in the credit-green relationship and, together
 556 with partial pooling across countries, gives more defensible inference than
 557 cluster-robust standard errors computed over only 41 clusters (Cameron and
 558 Miller, 2015). It carries two cross-level interactions: β_3 tests H2 directly, and
 559 β_4 tests whether regulatory quality on its own moderates the credit-green
 560 relationship. We are explicit that β_4 is not a test of H3. H3 asks whether

the SBFN moderation in H2 is itself conditional on regulatory quality, which is a three-way claim requiring a $\text{Credit}_{i,c} \times \text{SBFN}_c \times \text{RegQuality}_c$ term; this specification does not include one, and we do not add it here because the primary sample’s eight adopter economies cannot support a credibly estimated three-way interaction under partial pooling. H3 is instead tested directly by Stage 1’s fully saturated triple interaction above (Table 3, column M3) and, on an independent sample, by Stage 5’s no-fixed-effects specification (Section 5.7), both of which retain that term.

Stage 3 — causal forest / double machine learning. To avoid imposing a linear interaction functional form on how the treatment effect of credit access varies across the institutional landscape, we additionally estimate a Causal Forest with Double Machine Learning (Wager and Athey, 2018; Chernozhukov et al., 2018; Athey et al., 2019), using gradient-boosted nuisance models for the outcome and treatment propensity, and firm size, export status, SBFN status, and regulatory quality as effect-modifying features. This lets us recover conditional average treatment effects (CATEs)—the estimated effect of credit access on green adoption, separately for SBFN members versus non-members, and separately by regulatory-quality tercile—without imposing that the moderation take the single-coefficient interaction form assumed in Stages 1–2. We use “causal” here in the qualified sense standard in this literature: the forest’s identifying assumption is unconfoundedness conditional on the observed covariate set (firm size, export status, SBFN status, regulatory quality), not an exogenous source of variation in credit access itself. This is exactly the same assumption underlying the logit and hierarchical specifications; Stage 3 relaxes the functional form imposed on

586 that assumption from linear to fully flexible, but it does not relax the as-
587 sumption, and the CATEs reported below should be read accordingly as
588 doubly-robust conditional associations rather than causal effects identified
589 off a natural experiment.

590 **Stage 4 — small-sample supplementary check.** We additionally
591 re-estimate a reduced-form version of the credit-green relationship using a
592 waste-minimization-adoption outcome, available in four economies (India
593 2022, Indonesia 2023, Peru 2023, Timor-Leste 2021) whose Green Economy
594 modules asked this item under directly comparable wording. This outcome
595 is not part of the standardized cross-country database underlying Stage 5
596 below, so it offers genuinely independent, if small-sample, supplementary
597 evidence.

598 **Stage 5 — global-sample replication with country fixed effects.**
599 Finally, and centrally, we re-estimate the full specification on the 162-economy
600 global sample described in Section 4, using CO2-emissions monitoring as the
601 outcome. Because this sample spans far more country clusters than the
602 primary sample, we can now afford a country-fixed-effects specification: δ_c
603 absorbs SBFN_c and RegQuality_c 's main effects (collinear with the fixed effect
604 by construction, since both are constant within country-year) while leaving
605 their firm-level interactions with $\text{Credit}_{i,c}$ fully identified, since credit access
606 itself varies within country. This is a materially sharper test of H2 than Stage
607 1's clustered-SE approach, on an entirely independent sample and outcome
608 from Stages 1–3. The fixed-effects specification carries the two two-way in-
609 teractions only, so we report it alongside a no-fixed-effects specification for
610 two reasons: direct comparability with the primary sample's Table 3, and

because that specification, like Table 3's M3, retains the triple interaction
H3 requires.

Table 1: Variable definitions and sources.

Variable	Definition	WBES code	Source
green_adoption_binary	=1 if firm adopted ≥ 1 of 7 Green Economy practices in last 3 years	BMGc23b, c23d, c23e, c23f, c23j, c25, e5	WBES Green Economy Module (2018–2020)
fin_has_credit_line	=1 if firm has a line of credit or loan	K8	WBES Access to Finance Module
sbfn_member	=1 if SBFN policy adopted as of survey year	n/a	SBFN Global Progress Reports; data.sbfnetwork.org
wgi_regulatory_quality	WGI Regulatory Quality estimate, survey year	n/a	World Bank DataBank API (GOV_WGI_RQ_EST)
size_cat	Firm size stratum (Small/Medium/Large)	A6A	WBES Sampling Frame
sector_broad	Broad sector (Manufacturing/Services/Construction)	A4AX	WBES Sampling Frame
log_sales	Log of total annual sales	D2	WBES General Information Module
foreign_owned_dummy	=1 if any foreign ownership share $>0\%$	B2B	WBES Ownership Module
exporter_dummy	=1 if any export sales share $>0\%$	D3B, D3C	WBES Sales Module

5. Results

5.1. Descriptive patterns

Table 2: Sample composition, by economy.

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Albania 2019	377	2019	0	0.09	0.609	0.407
Armenia 2020	546	2020	0	0.21	0.684	0.482
Azerbaijan 2019	225	2019	0	-0.10	0.418	0.180
Belarus 2018	600	2018	0	-0.79	0.662	0.425
Bosnia and Herze- govina 2019	362	2019	0	-0.38	0.581	0.565
Bulgaria 2019	772	2019	0	0.46	0.604	0.467
Croatia 2019	404	2019	0	0.38	0.708	0.448
Cyprus 2019	240	2019	0	1.04	0.746	0.711
Czechia 2019	502	2019	0	1.12	0.725	0.526
Egypt 2020	3,075	2020	1	-0.64	0.303	0.065
Estonia 2019	360	2019	0	1.54	0.738	0.412
Georgia 2019	581	2019	1	0.88	0.441	0.501
Greece 2018	600	2018	0	0.29	0.825	0.499
Hungary 2019	805	2019	0	0.46	0.673	0.537
Italy 2019	760	2019	0	0.70	0.506	0.230
Jordan 2019	601	2019	1	0.16	0.471	0.248
Kazakhstan 2019	1,446	2019	0	0.11	0.519	0.257
Kosovo 2019	271	2019	0	-0.38	0.698	0.425
Kyrgyz Republic 2019	360	2019	1	-0.42	0.606	0.299
Latvia 2019	359	2019	0	1.00	0.824	0.386

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Lebanon 2019	532	2019	0	-0.56	0.452	0.365
Lithuania 2019	358	2019	0	0.98	0.678	0.342
Malta 2019	242	2019	0	0.83	0.901	0.479
Moldova 2019	360	2019	0	-0.11	0.619	0.328
Mongolia 2019	360	2019	1	-0.07	0.625	0.492
Montenegro 2019	150	2019	0	0.29	0.387	0.576
Morocco 2019	1,096	2019	1	0.03	0.478	0.251
North Macedonia 2019	360	2019	0	0.19	0.572	0.508
Poland 2019	1,369	2019	0	0.83	0.569	0.403
Portugal 2019	1,062	2019	0	1.00	0.812	0.600
Romania 2019	814	2019	0	0.36	0.438	0.474
Russia 2019	1,323	2019	0	-0.35	0.503	0.310
Serbia 2019	361	2019	0	0.06	0.654	0.585
Slovak Republic 2019	429	2019	0	0.73	0.782	0.386
Slovenia 2019	409	2019	0	0.88	0.793	0.674
Tajikistan 2019	352	2019	0	-0.80	0.523	0.194
Tunisia 2020	615	2020	1	-0.11	0.391	0.413
Türkiye 2019	1,663	2019	1	-0.02	0.346	0.345
Ukraine 2019	1,337	2019	0	-0.29	0.715	0.244
Uzbekistan 2019	1,239	2019	0	-0.92	0.751	0.240

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
West Bank and Gaza 2019	365	2019	0	-0.15	0.428	0.178

Note: Reg. qual. is the WGI Regulatory Quality estimate for the survey year. Green rate is the unconditional share of firms reporting adoption of at least one Green Economy practice. Credit rate is the share of firms reporting a bank credit line or loan.

615 Table 2 reports the full 41-economy sample composition; firm counts range
616 from 150 (Montenegro) to 3,075 (Egypt), and the raw, unconditional green-
617 practice-adoption rate ranges more than three-fold across economies, from
618 30.3% (Egypt) to 90.1% (Malta). Figure 1 maps this variation alongside
619 SBFN policy status: 8 of the 41 primary-sample economies had adopted
620 an SBFN framework as of their survey year (Georgia, Egypt, Jordan, Kyr-
621 gyz Republic, Mongolia, Morocco, Tunisia, and Türkiye)—13 of the full 47-
622 economy sample once the five (of six) SBFN-member small-sample supple-
623 mentary economies are counted—and every European Union member state in
624 the sample is, consistent with SBFN’s mandate, a non-adopter. At the firm
625 level, bank credit access correlates positively with the green-adoption index
626 ($r = 0.18$) but is essentially uncorrelated with a firm’s own perceived finance
627 obstacle ($r = -0.02$)—a first hint, well before any regression, that realized
628 access to credit rather than perceived financing difficulty is what tracks green
629 behavior, and the two are evidently not interchangeable measures of the same
630 underlying constraint.

631 Figure 2 previews the paper’s central finding graphically before a sin-
632 gle regression is estimated: plotting each economy’s green-adoption rate

633 against its regulatory-quality score and fitting separate lines for SBFN mem-
 634 bers and non-members produces two lines of very similar, modestly positive
 635 slope, separated by a roughly 15–20 percentage-point vertical gap—SBFN-
 636 member economies show systematically *lower* raw green-adoption rates at
 637 every level of regulatory quality, not a steeper relationship between regu-
 638 latory quality and adoption. That combination—a level difference with no
 639 slope difference—is exactly the signature the regression evidence below con-
 640 firms formally.

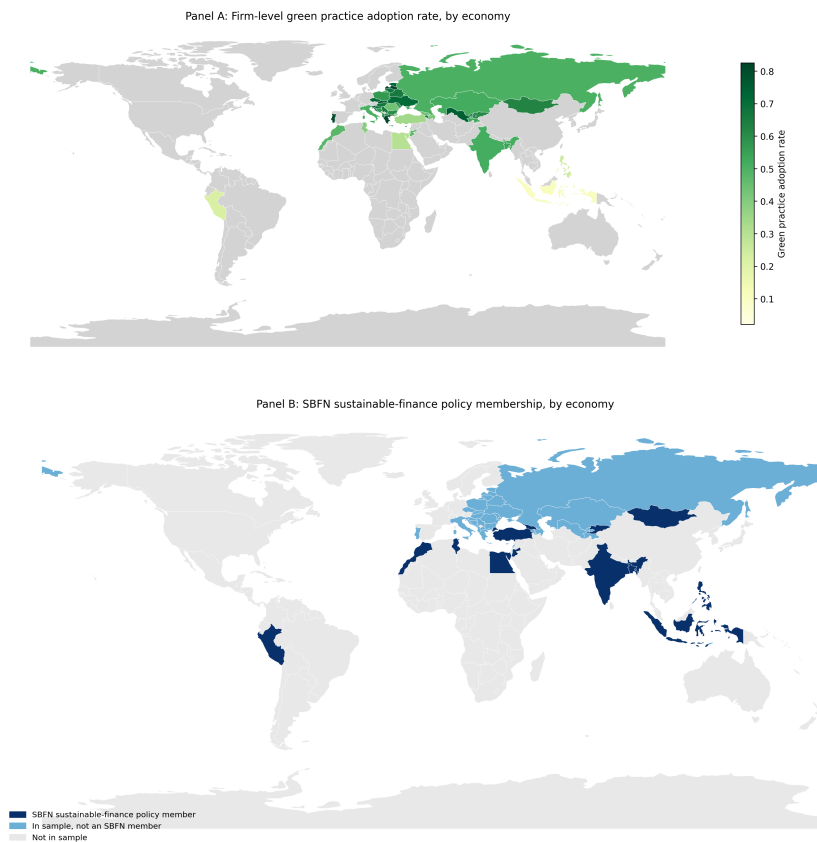


Figure 1: Green practice adoption and SBFN policy membership, by economy.

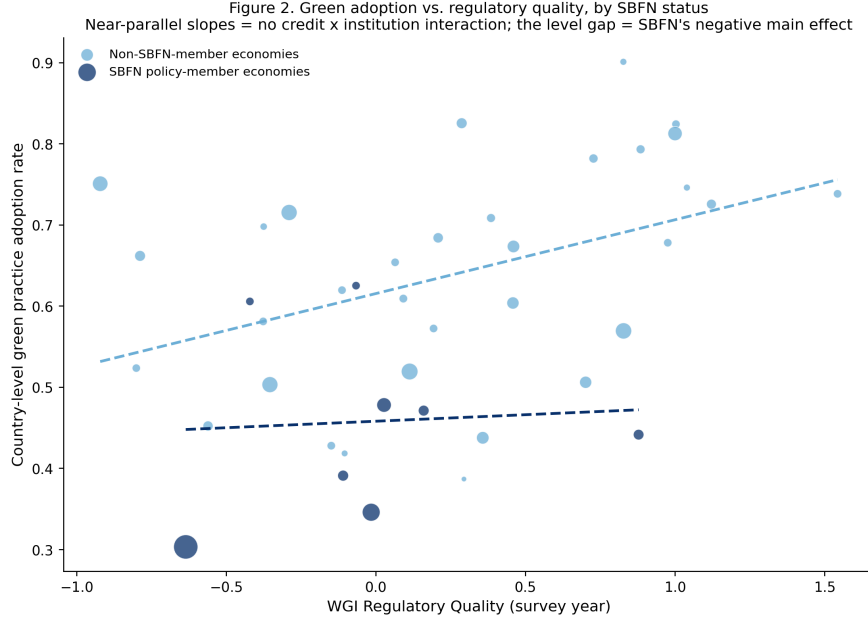


Figure 2: Green adoption vs. regulatory quality, by SBFN status.

5.2. Baseline classical benchmark

Table 3 reports the pooled logistic baseline. Column M1 confirms H1 on its own terms: firms with a bank credit line are significantly more likely to have adopted a green practice ($b = 0.571$, $p < 0.001$; average marginal effect = 12.95 percentage points, 95% CI [7.7, 18.2])—holding a formal credit line is associated with roughly the same increase in the probability of green adoption as moving from a small to a large firm, and a somewhat larger increase than being an exporter. Column M2 adds SBFN status and its interaction with credit access: the SBFN main effect is large, negative, and precisely estimated ($b = -0.991$, $p < 0.001$), while the credit $\times$ SBFN interaction is small and statistically indistinguishable from zero ($b = 0.049$, $p = 0.773$). Column M3 adds the fully saturated triple interaction with regulatory quality, and

653 it is this column, not the hierarchical model of Section 5.3, that carries the
654 paper’s direct test of H3: the credit $\times$ SBFN $\times$ regulatory-quality term is
655 small, wrong-signed relative to H3’s prediction, and nowhere near significance
656 ($b = -0.104$, $se = 0.272$, $p = 0.702$). The rest of M3 tells the same story
657 at greater length: every interaction term involving credit access is small
658 and insignificant, while regulatory quality’s uninteracted relationship with
659 adoption is positive, if only marginally significant in this particular fully-
660 saturated specification ($b = 0.278$, $p = 0.110$).

661 As set out in Section 4, adding country fixed effects here would absorb the
662 SBFN and regulatory-quality main effects entirely, while leaving the interac-
663 tion terms identified but reliant on the contrast between within-country credit
664 variation in this sample’s eight adopter economies and its thirty-three non-
665 adopters. Table 3 accordingly reports main effects and interactions jointly,
666 without country fixed effects; what it shows is a genuine test of H2 and, in
667 M3, of H3, not a placeholder—and that test returns no interaction on ei-
668 ther. The question the remaining two stages are built to answer is whether
669 that null result reflects a real absence of policy moderation, or an artifact
670 of forcing a two-level question through a single-level model with country-
671 clustered standard errors computed over only 41 clusters, a setting in which
672 such standard errors are known to be unreliable (Cameron and Miller, 2015).

673 5.3. Bayesian hierarchical model — primary test of H2

674 The hierarchical specification—country-varying random slopes on credit
675 access, cross-level interactions with SBFN status and standardized regulatory
676 quality, four chains of 1,000 post-warmup draws each, all $\hat{R} \leq 1.01$ and
677 ess_{bulk} between 797 and 2,270—is built specifically to give H2 a fair hearing

Table 3: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

	M1	M2	M3
Credit access	0.571*** (0.123)	0.459*** (0.090)	0.373*** (0.085)
SBFN member		-0.991*** (0.178)	-0.844*** (0.209)
Credit $\times$ SBFN		0.049 (0.168)	0.019 (0.140)
Regulatory quality			0.278 (0.174)
Credit $\times$ Reg. quality			0.144 (0.142)
SBFN $\times$ Reg. quality			0.053 (0.261)
Credit $\times$ SBFN $\times$ Reg. quality			-0.104 (0.272)
Log sales	0.061 (0.039)	0.041 (0.034)	0.068** (0.032)
Foreign owned	0.219 (0.143)	0.244* (0.139)	0.220 (0.143)
Exporter	0.537*** (0.092)	0.471*** (0.074)	0.412*** (0.079)
Sector, size controls	Yes	Yes	Yes
N	23,914	23,914	23,914
Pseudo R^2	0.058	0.089	0.093

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

under inference the 41-cluster classical specification cannot deliver, and to test regulatory quality’s own two-way moderation alongside it. H3’s three-way claim is not tested here (Section 4); it is tested in Table 3’s M3 above and in Section 5.7 below, and returns a null in both. This specification does not change the conclusion either. Table 4 reports the full posterior summary; three results stand out.

First, the firm-level credit effect (H1) survives unchanged in substance: posterior mean 0.33 (89% equal-tailed interval [0.18, 0.49]), comfortably excluding zero. Second, both country-level institutional variables retain clean, precisely estimated *main* effects on the baseline adoption rate—SBFN membership net-negative (-0.57, 89% ETI [-1.00, -0.09]), regulatory quality net-positive (0.34, 89% ETI [0.16, 0.53])—reproducing, with proper hierarchical uncertainty quantification rather than an absorbed or clustered approximation, exactly the level-shift pattern Figure 2 shows visually. Third, and centrally for H2, neither cross-level interaction is distinguishable from zero: credit $\times$ SBFN, 0.072 (89% ETI [-0.26, 0.40]); credit $\times$ regulatory quality (standardized), 0.004 (89% ETI [-0.13, 0.14])—a near-exact zero on the latter, not merely an imprecise one. The first of these is the direct H2 test; the second establishes that regulatory quality does not moderate the credit-green channel on its own, which is a necessary condition for the mechanism H3 posits without being a test of H3 itself.

We read this as an honest and, we think, informative finding rather than a disappointing one: giving the moderation hypothesis a hierarchical model built for precisely this two-level question still returns no interaction, while sharpening (relative to the pooled baseline) the confidence with which we

703 can state that SBFN status and regulatory quality shift the *level* of green
704 adoption directly.

Table 4: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

Parameter	Mean	SD	89% ETI low	89% ETI high	$\hat{R}$
Intercept	-1.93	0.246	-2.30	-1.50	1.00
Credit access	0.33	0.097	0.18	0.49	1.00
SBFN member	-0.57	0.300	-1.00	-0.09	1.01
Credit $\times$ SBFN	0.072	0.211	-0.26	0.40	1.00
Reg. quality (std.)	0.344	0.116	0.16	0.53	1.00
Credit $\times$ Reg. quality	0.004	0.087	-0.13	0.14	1.00

705 5.4. Causal forest / heterogeneous treatment effects

706 Table 5 and Figure 5 report the Causal Forest DML estimates, which im-
707 pose no linear-interaction functional form and let the treatment effect vary
708 flexibly across firm and country covariates. The overall average treatment
709 effect of credit access is 0.073 (95% CI [-0.040, 0.186])—directionally consis-
710 tent with, though smaller in magnitude and less precisely estimated than,
711 the logit average marginal effect of 0.130 in Section 5.2, reflecting the more
712 conservative variance properties of a doubly-robust, cross-fitted nonparamet-
713 ric estimator relative to a parametric logit on the same data. Critically,
714 the conditional average treatment effects triangulate with Section 5.3 rather
715 than contradicting it: the estimated effect is 0.072 for firms in non-SBFN
716 economies versus 0.076 for firms in SBFN economies, and 0.075 / 0.070 /

0.075 across the low/middle/high regulatory-quality terciles respectively—
three different institutional cuts, three essentially flat effects, all visibly overlapping in Figure 5. Two entirely independent estimators, one parametric and hierarchical, one nonparametric and forest-based, agree on the absence of institutional moderation.

The forest’s feature-importance decomposition adds a genuinely new result rather than only confirming the null, and speaks directly to the auxiliary H4 introduced in Section 3.4: of the four candidate effect-modifiers, firm size (log sales) accounts for 75.6% of the model’s explained heterogeneity, versus 19.9% for regulatory quality, 2.6% for SBFN status, and 1.9% for exporter status. Whatever heterogeneity exists in how strongly credit access predicts green adoption is, on this evidence, primarily a firm-level rather than a country-institutional phenomenon, supporting H4. Collapsing this into the Enterprise Surveys’ own coarse size strata, however, does not reveal a clean monotonic gradient: the estimated effect is 0.080 for small firms, 0.075 for medium firms, and 0.057 for large firms (Table 6; all with wide, overlapping confidence intervals spanning zero), which if anything runs opposite to a simple “larger firms better absorb a green investment’s fixed costs” story rather than confirming it. Read alongside the 75.6% importance share for the continuous size measure, we take this as evidence that the heterogeneity the forest is detecting is more granular or non-monotonic across the firm-size distribution than a three-category collapse can reveal, rather than evidence for any particular direction of a size gradient—a qualification we report explicitly rather than paper over with a tidier-sounding but unsupported claim. This redirects, rather than closes, the paper’s institutional-moderation question,

and we return to it in Section 7.

Table 5: Causal forest DML: average treatment effect, CATEs, and feature importances.

Estimate	Value	95% CI low	95% CI high
Overall ATE	0.0733	-0.0396	0.1862
CATE SBFN non-member	0.0723	-0.0300	0.1746
CATE SBFN member	0.0758	-0.0591	0.2107
CATE reg. quality tercile: low	0.0753	-0.0568	0.2075
CATE reg. quality tercile: mid	0.0697	-0.0355	0.1749
CATE reg. quality tercile: high	0.0749	-0.0203	0.1701
Feature	Importance		
SBFN member	0.0257		
Regulatory quality	0.1993		
Log sales	0.7562		
Exporter dummy	0.0189		

5.5. Small-sample supplementary check: waste-minimization adoption

Before turning to the global-sample replication that is this paper’s second major test (Section 5.7), Table 7 reports a smaller supplementary check using waste-minimization adoption — an outcome not captured in the standardized cross-country database underlying the global sample, and so a genuinely independent (if small-sample) data point — available in four economies whose Green Economy modules asked this item under directly comparable wording: India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021. Three of these four are themselves SBFN members at various progression stages, so

Table 6: Causal forest CATEs by firm-size category.

Size category	CATE	95% CI low	95% CI high
Small	0.0797	-0.0409	0.2004
Medium	0.0747	-0.0344	0.1838
Large	0.0569	-0.0428	0.1565

752 this sample cannot re-test the SBFN-adoption contrast; what it can test is
 753 whether the credit-to-green channel documented in Sections 5.2–5.4 is a pe-
 754 culiarity of the ECA-MENA sample or generalizes elsewhere. It generalizes:
 755 pooled across the four economies, having an overdraft facility is associated
 756 with significantly higher odds of waste-minimization adoption ($b = 0.422$,
 757 $p < 0.001$, $N = 9,038$, 4 countries). The direction of the core finance-green
 758 relationship replicates on a different continent, a different half-decade, and
 759 a narrower outcome definition than the one anchoring Sections 5.2–5.4 — a
 760 preview, on four economies, of the much larger replication in Section 5.7.

Table 7: Small-sample supplementary check: descriptive rates.

Economy	N	Waste-min. rate	Overdraft rate	SBFN member
India 2022	9,376	0.515	0.621	1
Indonesia 2023	2,955	0.336	0.144	1
Peru 2023	987	0.806	0.549	1
Timor-Leste 2021	238	0.061	0.089	0

761 5.6. Additional robustness

762 Table 9 subjects the primary-sample finding to four further checks, all
 763 retaining sector fixed effects, firm controls, and country-clustered standard

764 errors (no country fixed effects, for the reason given in Section 4). First,
 765 replacing the binary outcome with the continuous seven-item green-adoption
 766 index and estimating by OLS reproduces the same pattern to the decimal:
 767 credit access positive and significant ($b = 0.053$, $p < 0.001$), SBFN main
 768 effect negative and significant ($b = -0.092$, $p < 0.001$), interaction indistin-
 769 guishable from zero ($b = 0.006$, $p = 0.739$).

770 Second, replacing bank credit access with two alternative finance mea-
 771 sures produces one genuine point of nuance worth reporting rather than
 772 smoothing over. Using overdraft-facility access instead of a credit line/loan,
 773 the main overdraft effect remains positive and significant ($b = 0.274$, $p =$
 774 0.011), the SBFN main effect remains negative and significant ($b = -0.905$,
 775 $p < 0.001$)—but the overdraft $\times$ SBFN interaction is now negative and sig-
 776 nificant ($b = -0.346$, $p = 0.031$), the one interaction term in the entire paper
 777 that clears a conventional significance threshold, and it runs opposite to the
 778 direction H2 predicts. Using the reverse-coded perceived finance-obstacle
 779 measure in place of realized access, by contrast, both the main effect and
 780 its SBFN interaction are precisely zero ($b = -0.010$, $p = 0.827$; interac-
 781 tion $b = -0.012$, $p = 0.849$)—consistent with the descriptive correlation
 782 noted in Section 5.1. We read the overdraft result as suggestive rather than
 783 conclusive given it is one significant coefficient among more than a dozen
 784 interaction tests across the paper, but it is at least directionally consistent
 785 with a substantive story worth flagging for future work: if SBFN-guided
 786 lending is channelled preferentially through the term loans and credit lines
 787 that finance long-horizon capital expenditure, rather than through short-term
 788 overdraft facilities, we might expect exactly this pattern—a null-to-positive

789 interaction on credit lines and a negative one on overdrafts—though our
790 data cannot directly test whether SBFN-guided credit is disproportionately
791 term-structured.

792 Third, splitting the sample into manufacturing ($N = 13,340$) and services
793 ($N = 9,538$) subsamples shows the credit-access main effect, the SBFN main
794 effect, and the null credit $\times$ SBFN interaction all replicate within each sector
795 separately (manufacturing: credit $b = 0.480$, $p < 0.001$, interaction $b =$
796 -0.040 , $p = 0.826$; services: credit $b = 0.448$, $p < 0.001$, interaction $b =$
797 0.195 , $p = 0.426$)—the paper’s central null is not an artifact of pooling two
798 structurally different production technologies.

799 Because the overdraft interaction (the one coefficient in the paper that
800 clears a conventional significance threshold) is one of many credit $\times$ institutional-
801 moderator tests reported across the frequentist specifications in this paper,
802 we additionally apply a Benjamini–Hochberg false-discovery-rate correction
803 across the full family of 13 such coefficients reported under a classical (non-
804 Bayesian, non-forest) estimator in Tables 3, 9, and 10 (Table 8; full calcu-
805 lation reported in the replication package). The overdraft interaction’s raw
806 $p = 0.031$ becomes $p = 0.403$ after correction, and no coefficient in the
807 family survives at a false discovery rate of 0.10 or even 0.40. We read this
808 as reinforcing, with a formal multiple-testing accounting rather than only a
809 qualitative caveat, the suggestive-not-conclusive framing we already give this
810 result in Section 7: an isolated nominally-significant interaction among more
811 than a dozen tested is exactly what a pure false-discovery process would be
812 expected to produce roughly one time in twenty, and this correction does not
813 let us reject that account.

Table 8: Multiple-testing correction: Benjamini–Hochberg false discovery rate applied to every credit $\times$ institutional-moderator interaction coefficient reported under a frequentist estimator (Tables 3, 9, 10). Table 3 model M3 entries marked $\dagger$ are approximate p -values recovered from the reported coefficient and standard error via a normal approximation ($z = \text{coef}/\text{se}$), since the text reports only the coefficient and standard error for these terms.

Source	Term	Raw p	BH-adjusted p
Table 9 (b1) Overdraft	Overdraft $\times$ SBFN	0.031	0.403
Table 10 M2 (country FE)	Credit $\times$ SBFN	0.205	0.799
Table 10 M1 (no country FE)	Credit $\times$ Reg. quality †	0.301	0.799
Table 3 M3 †	Credit $\times$ Reg. quality	0.310	0.799
Table 10 M2 (country FE)	Credit $\times$ Reg. quality	0.346	0.799
Table 9 (c2) Services	Credit $\times$ SBFN	0.426	0.799
Table 10 M1 (no country FE)	Credit $\times$ SBFN	0.430	0.799
Table 3 M3 †	Credit $\times$ SBFN $\times$ Reg. quality	0.702	0.892
Table 9 (a) Continuous index, OLS	Credit $\times$ SBFN	0.739	0.892
Table 3 M2	Credit $\times$ SBFN	0.773	0.892
Table 9 (c1) Manufacturing	Credit $\times$ SBFN	0.826	0.892
Table 9 (b2) Finance obstacle (reverse-coded)	Obstacle $\times$ SBFN	0.849	0.892
Table 3 M3 †	Credit $\times$ SBFN	0.892	0.892

Table 9: Additional robustness checks.

Specification	Finance coef. (p)	SBFN coef. (p)	Interaction coef. (p)
(a) Continuous index, OLS	0.053 (0.000)	-0.092 (0.000)	0.006 (0.739)
(b1) Overdraft	0.274 (0.011)	-0.905 (0.000)	-0.346 (0.031)
(b2) Finance obstacle (reverse-coded)	-0.010 (0.827)	-1.011 (0.000)	-0.012 (0.849)
(c1) Manufacturing subsample	0.480 (0.000)	-1.047 (0.000)	-0.040 (0.826)
(c2) Services subsample	0.448 (0.000)	-0.914 (0.001)	0.195 (0.426)

814 5.7. *Global-sample replication with country fixed effects*

815 The preceding five subsections establish the paper’s central finding on the
816 primary sample. We now subject that finding to what we consider the most
817 demanding test available to us: complete re-estimation on an independently
818 assembled, 162-economy global sample (Section 4), using a different outcome
819 variable (CO2-emissions monitoring rather than the seven-item composite),
820 different survey years (2021–2026 rather than 2018–2020), and, for the first
821 time in the paper, a country-fixed-effects specification that this much larger
822 set of clusters can support. Figure 3 maps the combined reach of the two
823 samples together: 166 distinct economies, spanning every region and, through
824 the global sample specifically, extending for the first time in this literature
825 into high-income economies with no SBFN involvement at all.

826 Table 10 reports two specifications. Column M1, without country fixed
827 effects, is the direct global-sample analogue of Table 3’s primary-sample base-
828 line: credit access remains positive and significant ($b = 0.198$, $p = 0.039$,
829 $N = 89,797$, 159 countries), while the SBFN main effect ($b = -0.171$,
830 $p = 0.353$), the credit $\times$ SBFN interaction ($b = 0.278$, $p = 0.430$), and
831 every regulatory-quality interaction are statistically indistinguishable from
832 zero. Like Table 3’s M3, this specification is fully saturated and so retains
833 the credit $\times$ SBFN $\times$ regulatory-quality term H3 requires, which is likewise
834 insignificant; Table 10 tabulates the focal two-way terms only. H3 therefore
835 receives a direct, fully saturated test on both firm-level samples, and returns a
836 null on each. Column M2 imposes country fixed effects, absorbing the SBFN
837 and regulatory-quality main effects by construction (they are collinear with
838 δ_c , since both are constant within a given country-year) while leaving their

839 firm-level interactions with credit access fully identified, because credit access
840 itself varies within country. This is a sharper test than anything the primary
841 sample’s single 2018–2020 wave can support, and it returns the same answer
842 with even more precision: credit access is strongly significant ($b = 0.249$,
843 $p < 0.001$), while the credit $\times$ SBFN interaction ($b = 0.506$, $p = 0.205$) and
844 the credit $\times$ regulatory-quality interaction ($b = -0.044$, $p = 0.346$) remain
845 null.

846 We regard this as the paper’s most important robustness result. It is not
847 a re-analysis of the same data under a different label: it is a different outcome
848 construct, a different set of economies (only a handful of which overlap with
849 the primary sample), a different half-decade, and an econometric specification
850 — country fixed effects — that is only available at all because this second
851 sample has enough clusters to support it. That the qualitative conclusion
852 is identical across both samples and across the full set of four estimators
853 reported in this paper (classical logit, Bayesian hierarchical, causal forest, and
854 now country-fixed-effects logit) is, we think, considerably stronger evidence
855 for the paper’s central claim than either sample could offer alone.

856 *5.8. Results summary across all specifications*

857 Sections 5.2–5.7 report ten tables spanning four estimators and two inde-
858 pendent samples. Figure 4 consolidates every coefficient bearing on H1–H3
859 into a single view. Panel A shows the credit-access effect (H1): every point
860 estimate is positive and every confidence interval excludes zero, in both sam-
861 ples and all four estimators. Panels B and C show the credit $\times$ SBFN and
862 credit $\times$ regulatory-quality interactions (H2, H3): every single confidence in-
863 terval, across both samples and every estimator, straddles zero. Figure 5

Table 10: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

	M1: no country FE	M2: country FE
Credit access	0.198** (0.096)	0.249*** (0.057)
SBFN member	-0.171 (0.184)	— (absorbed)
Credit $\times$ SBFN	0.278 (0.352)	0.506 (0.399)
Regulatory quality (std.)	0.155 (0.096)	— (absorbed)
Credit $\times$ Reg. quality	0.092 (0.089)	-0.044 (0.047)
Log sales	0.127*** (0.027)	0.288*** (0.028)
Foreign owned	0.621*** (0.083)	0.528*** (0.046)
Exporter	0.456*** (0.094)	0.335*** (0.044)
Country fixed effects	No	Yes
N	89,797	89,797
Countries	159	159

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Taiwan dropped (no WGI coverage).

Figure 3. Combined sample coverage: 166 economies (41 primary rich-module + 162 global minimal-indicator, net of overlap)

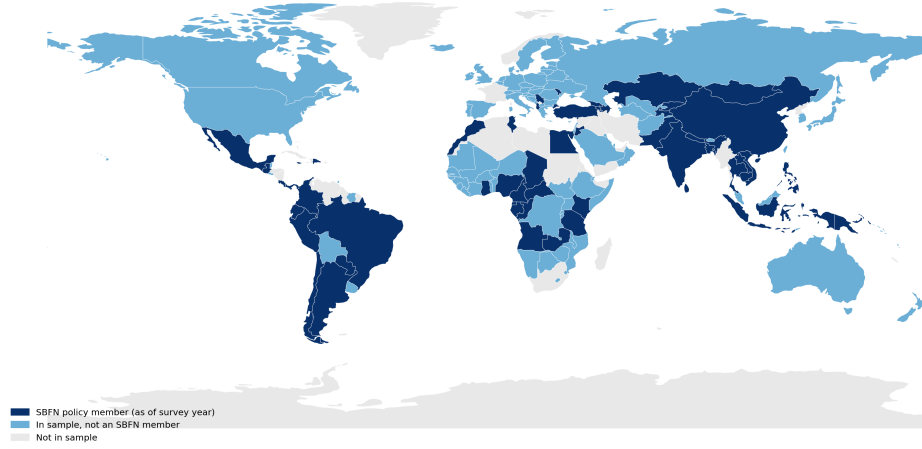


Figure 3: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

864 shows the same non-result from a completely different angle—the causal
 865 forest’s conditional average treatment effects broken out by SBFN status,
 866 regulatory-quality tercile, and firm-size category all overlap both each other
 867 and the overall average treatment effect. No subgroup split, by any variable
 868 examined in this paper, recovers a distinguishable effect.

Figure 4. Coefficient estimates across every specification and both samples
 (Panel A: robust and significant everywhere; Panels B-C: null everywhere)

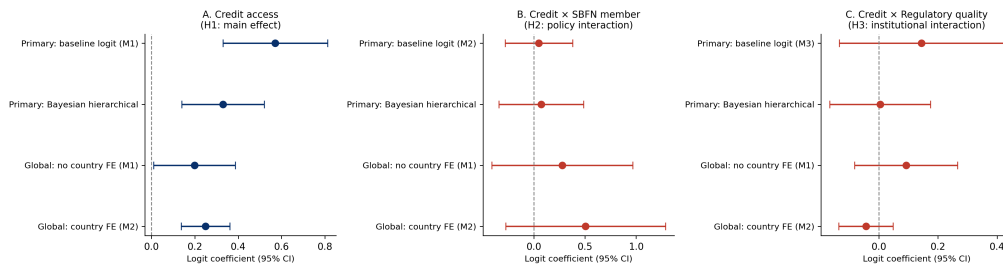


Figure 4: Coefficient estimates across every specification and both samples.

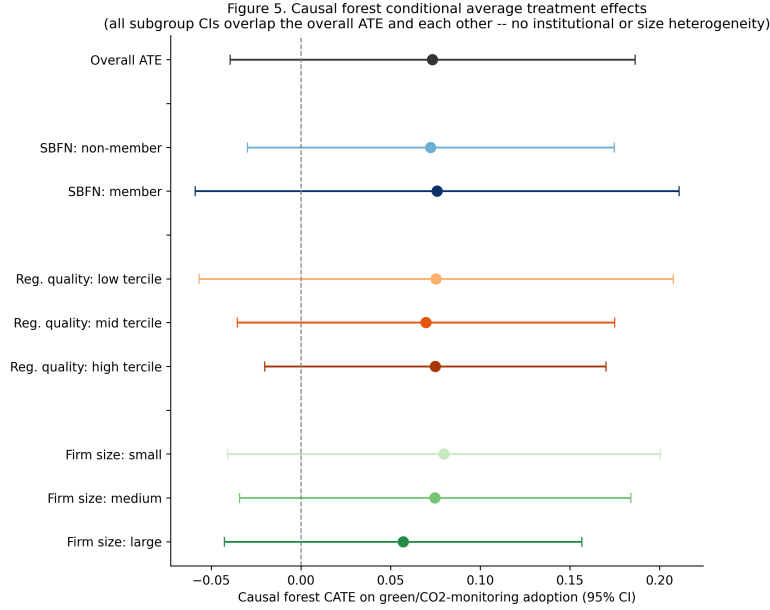


Figure 5: Causal forest conditional average treatment effects, by subgroup.

6. Supplementary evidence: a country-level staggered event-study on macro environmental outcomes

Sections 5.2–5.8 establish this paper’s central finding using firm-level cross-sectional data, for the data-availability reasons given in Section 4: the WBES Green Economy module cannot support a staggered difference-in-differences or event-study design around each economy’s own SBFN adoption date, because it has not been fielded as a repeated panel instrument. That constraint applies to the firm-level finance-to-green-investment channel specifically; it does not apply to a different, complementary question that the same SBFN adoption-timing variation can answer directly: does a country’s aggregate environmental trajectory shift around its own SBFN adoption date? This section builds and reports that test, using an entirely indepen-

881 dent data source and unit of analysis, precisely because it is the identification
882 strategy this literature most often asks for, and because a genuine answer,
883 however it comes out, is more useful than leaving the question unaddressed.

884 *6.1. Data and design*

885 We assemble a country-year panel spanning every World Bank member
886 economy with a non-aggregate ISO3 code (217 economies), 2000–2024. Treat-
887 ment timing is each economy’s SBFN join year, drawn from the same SBFN
888 data portal membership roster used to construct the global sample (Sec-
889 tion 4) and cross-checked against the global-sample SBFN coding used in
890 Section 5.7 (zero disagreements across 61 overlapping economies); economies
891 that have never joined SBFN, as of the roster’s most recent update, serve as
892 the never-treated comparison group. This yields 68 ever-treated economies
893 (join years 2012–2024, the same real staggered-adoption variation described
894 in Section 2) and 149 never-treated economies. Outcome data come from
895 the World Bank’s World Development Indicators, retrieved via the World
896 Bank DataBank API: renewable energy consumption (% of total final en-
897 ergy consumption, `EG.FEC.RNEW.ZS`) and CO2 emissions per capita (AR5
898 GWP-consistent series, `EN.GHG.CO2.PC.CE.AR5`), both available for the large
899 majority of economies from 2000 through at least 2021–2024. We addition-
900 ally draw on GDP per capita (constant 2015 US\$, `NY.GDP.PCAP.KD`) as a
901 covariate, to net out the obvious confound that SBFN-adopting economies
902 are disproportionately still-industrializing, faster-growing economies whose
903 emissions profiles are shaped by their development stage independently of
904 any banking-sector policy.

905 We estimate the effect of SBFN adoption on each outcome using the

906 Callaway and Sant’Anna (2021) staggered-adoption difference-in-differences
 907 estimator (Callaway and Sant’Anna, 2021), which recovers group-time av-
 908 erage treatment effects for each adoption cohort separately and aggregates
 909 them into an overall average treatment effect on the treated (ATT) and
 910 an event-study profile. We report this estimator, rather than a standard
 911 two-way-fixed-effects (TWFE) regression, because TWFE is now well known
 912 to produce badly biased estimates under staggered treatment timing when
 913 already-treated units are implicitly used as controls for later-treated units
 914 (Goodman-Bacon, 2021); we report a naive TWFE specification alongside
 915 the Callaway-Sant’Anna estimates specifically to illustrate the size of that
 916 bias in our own data, rather than as a candidate preferred estimate. For
 917 each outcome we report both an unconditional specification and a doubly-
 918 robust specification conditioning on log GDP per capita, and we check the
 919 doubly-robust specification’s sensitivity to the choice of comparison group,
 920 using never-treated economies as our preferred comparison group throughout
 921 and not-yet-treated economies (those that will eventually adopt but had not
 922 yet, as of the relevant relative period) as a robustness check.

923 *6.2. Results*

924 Table 11 reports the overall ATT for both outcomes across four specifica-
 925 tions, including a comparison-group robustness check; Figure 6 plots the full
 926 event-study profile (doubly-robust, GDP-per-capita-controlled, never-treated
 927 specification) for both outcomes.

928 Two results, read together, do not support treating SBFN adoption as a
 929 country-level lever that visibly improves aggregate environmental outcomes –
 930 which is itself informative, since this is the identification strategy best suited

Table 11: Country-level staggered event-study: overall average treatment effect of SBFN adoption on macro environmental outcomes. Callaway-Sant’Anna estimator, 999 bootstrap iterations; naive TWFE reported for comparison only. Full country-year panel, 2000–2024.

Outcome	Specification	ATT	SE	95% CI
Renewable energy %	Callaway-Sant’Anna, no covariates, never-treated	-1.192	0.802	[-2.763, 0.380]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.674	0.916	[-1.122, 2.470]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.653	0.919	[-1.147, 2.454]
Renewable energy %	Naive TWFE (post dummy, country+year FE)	-4.672***	1.127	[-6.882, -2.463]
CO2 per capita	Callaway-Sant’Anna, no covariates, never-treated	0.631***	0.179	[0.280, 0.982]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.349**	0.156	[0.044, 0.655]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.313**	0.148	[0.023, 0.603]
CO2 per capita	Naive TWFE (post dummy, country+year FE)	0.901***	0.204	[0.500, 1.302]

*** 95% CI excludes zero; ** 95% CI excludes zero, narrower margin. Underlying data and code in the replication package.

Country-level staggered event-study, SBFN adoption on macro environmental outcomes
(Callaway-Sant'Anna, doubly-robust, log GDP per capita control; shaded = pointwise 95% CI;
relative periods beyond $\pm 12/9$ trimmed, too few surviving cohorts to be informative)

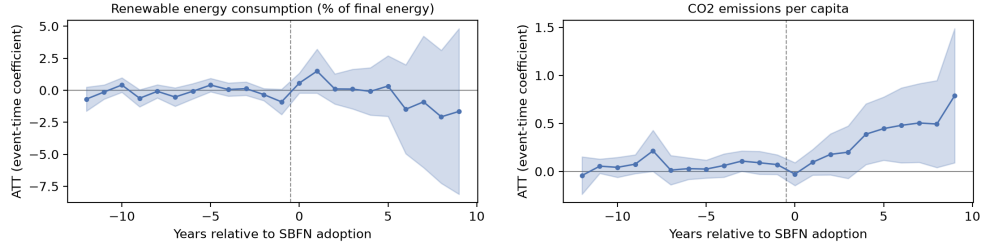


Figure 6: Country-level staggered event-study, doubly-robust specification with log GDP per capita control. Shaded bands are pointwise 95% confidence intervals; relative periods beyond $\pm 12/9$ are trimmed for having too few surviving cohorts to be informative.

931 to answer that specific question directly. First, renewable energy share shows
 932 no effect distinguishable from zero in either specification, and the doubly-
 933 robust event-study profile (Figure 6, left panel) is flat and precisely estimated
 934 on both sides of the adoption date: a clean null, not an underpowered one.

935 Second, CO2 emissions per capita shows a positive (i.e., the “wrong-
 936 signed” direction if SBFN adoption reduced emissions) and statistically sig-
 937 nificant association with adoption in every specification, but three features
 938 of the result argue against reading it causally. It shrinks by roughly 45%
 939 (0.631 to 0.349) once we condition on log GDP per capita, indicating that
 940 a substantial share of the raw association is attributable to exactly the
 941 development-stage confound we set out to net out. The doubly-robust event-
 942 study profile (Figure 6, right panel) shows a small, mostly flat pre-period
 943 with only two marginally significant leads, followed by a post-period effect
 944 that grows smoothly and monotonically from near zero at adoption to its
 945 largest magnitude nine to twelve years out – a continuing-divergence pattern

946 more consistent with SBFN-adopting economies already sitting on a different,
947 faster-rising emissions trajectory than non-adopters (which a single covariate
948 cannot fully absorb) than with a discrete policy-caused jump at the adop-
949 tion date. And the naive TWFE estimate (0.901) is itself more than double
950 the doubly-robust Callaway-Sant’Anna estimate (0.349), the same direction
951 of bias documented in the staggered-DID literature (Goodman-Bacon, 2021)
952 and a useful internal illustration of why we lead with the more robust esti-
953 mator throughout this section.

954 Both doubly-robust results are stable to the choice of comparison group:
955 substituting not-yet-treated economies for never-treated economies leaves the
956 CO2 estimate essentially unchanged (0.313 versus 0.349, both significant at
957 conventional levels) and the renewable-energy null equally unchanged (0.653
958 versus 0.674, neither distinguishable from zero); Table 11 reports both. Nei-
959 ther conclusion in this section depends on which set of untreated economies
960 serves as the counterfactual.

961 Neither result supports a “green banking policy backfires” reading, and
962 neither supports a “green banking policy works” one. The more measured
963 reading is that a macro-level design – built using exactly the identification
964 strategy (staggered adoption timing, a proper Callaway-Sant’Anna estima-
965 tor, never-treated controls) this literature’s own methodological expectations
966 call for – also does not produce clean evidence that SBFN adoption changes
967 a country’s aggregate environmental trajectory for the better, and the one
968 statistically significant result we do find is more parsimoniously explained by
969 which economies self-select into SBFN membership than by what the policy
970 itself does once adopted. This is, we think, a genuinely different and com-

971plementary piece of evidence to Sections 5.2–5.7’s firm-level results, not a
972restatement of them: different data source, different unit of analysis, differ-
973ent outcome constructs, and, for the first time in this paper, an identification
974strategy that exploits exogenous variation in adoption *timing* rather than re-
975lying on cross-sectional comparison across adopters and non-adopters. That
976it converges on the same substantive conclusion – no clean evidence of a ben-
977eficial institutional effect – from a design built specifically to meet a bar the
978paper’s firm-level evidence cannot meet, strengthens rather than substitutes
979for the paper’s central finding.

980 7. Discussion

981Put in plain economic terms, the paper’s central finding is this: a firm
982holding a bank credit line is associated with an adoption probability roughly
98313 percentage points higher than an otherwise-similar firm without one—an
984association net of observed controls, not a causally identified effect (Limita-
985tion 1)—and that gap is essentially unchanged whether the firm sits in an
986SBFN adopter or non-adopter, or across the full range of regulatory qual-
987ity (Figure 4). Firms in SBFN-adopting economies are, on average, 15–20
988percentage points *less* likely to have adopted a green practice at all—a level
989gap, not a slope gap.

990This convergence is informative precisely because of how differently the
991evidence could have failed to agree. The primary and global samples share al-
992most no economies, span different half-decades, use different outcome items,
993and reach institutional settings—high-income, non-SBFN economies such as
994Australia and Germany—the primary sample cannot touch. Section 6’s

995 country-level staggered event-study adds a third, independent test on top
 996 of this: a different data source, a different unit of analysis, and the paper’s
 997 only design that exploits exogenous variation in adoption *timing*. That it
 998 too finds no clean evidence of a beneficial institutional effect—a precise null
 999 on renewable-energy share, and a CO2 association that shrinks substantially
 1000 once development stage is controlled for—is the hardest-to-dismiss evidence
 1001 that this is a real feature of the data, not an artifact of any one sample, level
 1002 of analysis, or estimator. (This is a distinct claim from H2/H3’s firm-level
 1003 moderation null—a main effect at country level, not a slope change at firm
 1004 level—but evidence of the same kind.)

1005 This complicates rather than confirms the implicit premise of a green-
 1006 credit-policy literature disproportionately built on China’s unusually mature,
 1007 well-resourced guidelines (Li et al., 2023; Huang et al., 2023; Dai et al., 2025).
 1008 Our sample spans every stage of SBFN’s own progression framework, from
 1009 Jordan and the Kyrgyz Republic’s initial “Commitment” to Mongolia’s “Ad-
 1010 vancing,” and across that range we find no evidence that formal adoption
 1011 changes how credit access translates into green investment. The causal for-
 1012 est’s feature-importance decomposition (Section 5.4) offers one redirection
 1013 rather than a tidy answer: the heterogeneity that does exist is overwhelm-
 1014 ingly a firm-size phenomenon (75.6% of explained variance) rather than in-
 1015 stitutional (regulatory quality 19.9%, SBFN 2.6%)—though it does not re-
 1016 solve into a clean size gradient, so we can say where the heterogeneity sits
 1017 without yet saying in which direction it runs. If the binding constraint is
 1018 which firms can absorb a green investment’s fixed costs once financed, rather
 1019 than whether a supervisory guideline nudges banks to lend more broadly, a

1020 country-level mandate on the extensive margin of lending may be aimed at
1021 the wrong margin regardless of its own credibility.

1022 “No interaction detected” is not the same claim as “no institutional com-
1023 plementarity exists,” and we do not want the former to be read as settling
1024 the latter. Two alternative readings stay live: a linear interaction may be
1025 the wrong functional form for a genuinely threshold-shaped complementarity,
1026 given SBFN’s own graded progression framework and our binary coding of
1027 it (Limitation 2); and the Worldwide Governance Indicators may simply not
1028 proxy the banking-supervisory capacity a green-lending guideline depends
1029 on (Limitation 5). We think our preferred reading—no complementarity
1030 through this channel—is best supported by the evidence, but hold it as the
1031 best current account rather than a claim that rules the alternatives out.

1032 Two further limits are worth naming directly. First, SBFN guidelines
1033 could in principle be working on the *extensive* margin of who obtains credit
1034 at all, rather than the *intensive* margin our interaction terms test; testing
1035 that would require modeling credit access itself as a function of SBFN status,
1036 which our demand-side design does not do. Second, every outcome in this
1037 paper is firm-reported, and we cannot rule out a compliance-reporting-versus-
1038 practice gap between what firms report and what banks report to their own
1039 supervisor—a limit no firm-level survey instrument can fully resolve on its
1040 own. Read together with Section 6, whose null is, to our knowledge, the first
1041 test of the SBFN framework’s own theory of change against outcomes on
1042 the ground rather than its own tracked policy-stage progression, both point
1043 the same way: a framework can look like it is working by every measure the
1044 network itself tracks while the channel it is meant to move remains, on this

1045 evidence, unmoved.

1046 The one interaction that clears a conventional significance threshold—a
1047 negative credit $\times$ SBFN effect for overdraft rather than term-credit access
1048 (Section 5)—is more consistent with this redirection than with a simple policy
1049 failure: overdrafts finance working capital, not the long-horizon expenditure
1050 our outcome measures, so a supervisor successfully steering credit toward
1051 term instruments would produce exactly this pattern. We flag it as suggestive
1052 only: it is one significant coefficient among many tested, and does not survive
1053 a Benjamini–Hochberg correction at any conventional false discovery rate
1054 (Table 8).

1055 Read against the broader themes of state capacity and policy credibility,
1056 our results sit closer to a state-capacity-is-necessary-but-firm-level-frictions-
1057 bind-independently story than a straightforward complementarity one. Reg-
1058 ulatory quality raises the baseline propensity to adopt green practices, but
1059 is not the missing ingredient that lets a green banking guideline reshape
1060 the credit-to-green channel specifically. The binding constraint looks less
1061 like whether the supervising jurisdiction can make its guideline credible, and
1062 more like whether firms below a certain scale can absorb a green investment’s
1063 fixed costs once credit is available—a question one level down from the one
1064 the SBFN framework is designed to answer.

1065 **8. Conclusion**

1066 This paper set out to test whether the wave of green banking policy
1067 adoption coordinated through the Sustainable Banking and Finance Net-
1068 work changes the relationship between firm-level access to finance and green

1069 investment behavior, and whether that depends on the regulatory capacity
 1070 available to make the policy credible. Using harmonized WBES data across
 1071 41 economies (2018–2020), a four-country waste-minimization check, and an
 1072 independently assembled 162-economy global sample (2021–2026) reaching
 1073 for the first time into high-income, non-SBFN economies, we apply four es-
 1074 calating estimators and find five consistent results. First, bank credit access
 1075 is robustly associated with green practice adoption, surviving every specifi-
 1076 cation, subsample, and alternative-outcome check, and replicating in both
 1077 the small waste-minimization sample and the full global sample. Second,
 1078 neither SBFN adoption nor regulatory quality moderates that relationship
 1079 under any estimator, on either sample, though both shift the general level of
 1080 adoption directly in the primary sample. Third, where effect heterogeneity
 1081 does exist, it sits at the level of firm size rather than country institutions.
 1082 Fourth, this pattern replicates on a sample sharing almost no economies with
 1083 the primary one, under a materially sharper identification strategy. Fifth,
 1084 and most importantly for the paper’s credibility, an independent country-
 1085 level staggered event-study (Section 6)—a main-effect test, not a repeat of
 1086 the firm-level moderation null—likewise finds no clean evidence that SBFN
 1087 adoption improves aggregate environmental outcomes, the single strongest
 1088 evidence that this null is a real feature of the data rather than an artifact of
 1089 any one sample, data source, level of analysis, or estimator.

1090 For policy, the implication is not that green banking guidelines are in-
 1091 effective, but that our evidence does not support treating them as a lever
 1092 for widening the green-investment gap between financed and unfinanced
 1093 firms—the margin most of this literature implicitly assumes they operate

1094 on. If our tentative loan-tenor reading of the overdraft result is correct,
1095 policymakers may find more traction focusing on the term-lending instru-
1096 ments that actually finance green capital expenditure, and on complementary
1097 measures—technical assistance, matching grants, credit guarantees—that ad-
1098 dress smaller firms’ absorptive-capacity constraints directly.

1099 We close with seven limitations. First, both firm-level samples are single
1100 cross-sections (Section 4 explains why a genuine panel does not exist for this
1101 outcome): they cannot rule out reverse causality or unobserved country char-
1102 acteristics correlated with both SBFN adoption and baseline green-adoption
1103 propensity, and the global sample’s fixed effects sharpen this only for the
1104 interaction terms, not the credit main effect. Second, SBFN membership
1105 is a binary proxy for what the network’s own framework treats as a graded
1106 process; the global sample’s regional roster entries (the CEMAC and East-
1107 ern Caribbean groupings, Section 4) additionally rest on an inference about
1108 coverage rather than an explicit per-country statement. Third, the four-
1109 economy waste-minimization check cannot re-test the SBFN contrast (three
1110 of its four economies are members) and combines non-identical outcome items
1111 by construction of WBES’s own shortened module design—a data-availability
1112 constraint, not a methodological choice. Fourth, the global sample’s CO2-
1113 monitoring outcome is a single item, narrower than the primary sample’s
1114 seven-item composite; that the same pattern holds on it is reassuring but
1115 does not establish it would hold on a richer measure. Fifth, the World-
1116 wide Governance Indicators measure is a broad governance indicator, not a
1117 banking-supervision-specific capacity measure. Sixth, our design tests only
1118 the intensive margin of the credit-to-green channel and cannot rule out that

1119 SBFN policy operates on the extensive margin of who obtains credit, nor dis-
1120 tinguish firm-reported practice adoption from a bank-compliance-reporting
1121 gap (Section 7). Seventh, Section 6’s event-study is an ecological-level test
1122 of a related but distinct question; its one significant result (CO2 per capita)
1123 is netted against a single covariate, and we cannot rule out that a richer set
1124 of development-stage controls would shrink it further, or that some genuine,
1125 small, confounded relationship remains. We leave each of these to future
1126 work.

1127 **Declaration of generative AI and AI-assisted technologies in the** 1128 **manuscript preparation process**

1129 During the preparation of this work the author(s) used AI-assisted tools
1130 to help polish the writing. After using these tools, the author(s) reviewed
1131 and edited the content as needed and take full responsibility for the content
1132 of the published article.

1133 **Appendix A. SBFN policy status and regulatory quality, full sam-** 1134 **ple**

1135 Table A.12 reports, for all 47 economy-years in the paper’s primary and
1136 small-sample supplementary check, the SBFN membership status coded for
1137 the analysis, the join year where applicable, the SBFN Multi-Tiered Progress
1138 Framework stage in force nearest the survey year, and the exact year of the
1139 Worldwide Governance Indicators observation used. Full per-country source
1140 citations (the specific SBFN Global Progress Report, country addendum, or
1141 accession announcement consulted for every determination) are available in

the online supplementary material. The analogous 162-economy table for the global sample (Section 4) is summarized by region in Appendix C and provided in full in the online supplementary material and the replication package.

Table A.12: SBFN policy status and regulatory quality, full 47-economy sample.

Economy	SBFN member	Join year	Stage at survey	WGI year
Albania 2019	0	—	None	2019
Armenia 2020	0	—	None	2020
Azerbaijan 2019	0	—	None	2019
Bangladesh 2022	1	2012	Advancing (ESG Integration)	2022
Belarus 2018	0	—	None	2018
Bosnia and Herzegovina 2019	0	—	None	2019
Bulgaria 2019	0	—	None	2019
Croatia 2019	0	—	None	2019
Cyprus 2019	0	—	None	2019
Czechia 2019	0	—	None	2019
Egypt 2020	1	2016	Formulating (Preparation)	2020
Estonia 2019	0	—	None	2019
Georgia 2019	1	2017	Developing (Implementation)	2019
Greece 2018	0	—	None	2018
Hungary 2019	0	—	None	2019

Economy	SBFN member	Join year	Stage at survey		WGI year
India 2022	1	2016	Developing	(Commitment)	2022
Indonesia 2023	1	2012	Consolidating (Maturing)		2023
Italy 2019	0	—	None		2019
Jordan 2019	1	2016	Commitment	(Preparation)	2019
Kazakhstan 2019	0	—	None		2019
Kosovo 2019	0	—	None		2019
Kyrgyz Republic 2019	1	2018	Commitment	(Preparation)	2019
Latvia 2019	0	—	None		2019
Lebanon 2019	0	—	None		2019
Lithuania 2019	0	—	None		2019
Malta 2019	0	—	None		2019
Moldova 2019	0	—	None		2019
Mongolia 2019	1	2012	Advancing	(Implementation)	2019
Montenegro 2019	0	—	None		2019
Morocco 2019	1	2014	Advancing	(Implementation)	2019
North Macedonia 2019	0	—	None		2019
Peru 2023	1	2013	Advancing	(ESG Integration)	2023
Philippines 2023	1	2013	Advancing	(ESG Integration)	2023
Poland 2019	0	—	None		2019

Economy	SBFN member	Join year	Stage at survey	WGI year
Portugal 2019	0	—	None	2019
Romania 2019	0	—	None	2019
Russia 2019	0	—	None	2019
Serbia 2019	0	—	None	2019
Slovak Republic 2019	0	—	None	2019
Slovenia 2019	0	—	None	2019
Tajikistan 2019	0	—	None	2019
Timor-Leste 2021	0	—	None	2021
Tunisia 2020	1	2019	Unverified	2020
Türkiye 2019	1	2015	Developing (Implementation)	2019
Ukraine 2019	0	—	None	2019
Uzbekistan 2019	0	—	None	2019
West Bank and Gaza 2019	0	—	None	2019

1146 Appendix B. Bayesian model convergence diagnostics

1147 Convergence diagnostics for every focal parameter of the hierarchical
1148 model reported in Table 4 are included directly in that table ($\hat{R}$ column); the
1149 full posterior summary also includes ess_{bulk} ranging from 797 (SBFN member)
1150 to 2,270 (credit $\times$ regulatory quality), comfortably above the conventional
1151 minimum threshold of 400 effective samples per parameter for stable poste-
1152 rior summaries at four chains. No divergent transitions were flagged during

1153 sampling (NUTS via the nutpie sampler, four chains of 1,000 post-warmup
1154 draws each after 1,000 tuning iterations).

1155 **Appendix C. Global-sample SBFN status, regional summary**

1156 Of the 162 global-sample economy-years, 61 are coded as SBFN members
1157 as of their survey year and 101 as non-members. Table C.13 reports the
1158 regional distribution of the 61 members, drawn from the SBFN data por-
1159 tal’s regional classification; five of the 61 (Peru, Philippines, Timor-Leste,
1160 Bangladesh, and Indonesia) carry the deeper per-country sourcing already
1161 reported in Appendix A, having been determined during the primary sam-
1162 ple’s original research pass rather than from the portal roster alone.

Table C.13: Global-sample SBFN members, by SBFN region.

Region	Member economy-years
Latin America and Caribbean	17
Africa	12
Europe and Central Asia	11
East Asia and Pacific	11
South Asia	6
Middle East and North Africa	4
Total	61

1163 **Additional material**

1164 *Online appendix*

1165 Appendix A, Appendix B, and Appendix C appear above; per-country
1166 source citations at the level of detail compiled during data construction (in-
1167 dividual SBFN Global Progress Report page references, country-addendum
1168 URLs, and IFC accession-announcement links for every non-obvious deter-
1169 mination in the primary sample; the full 162-row global-sample SBFN/WGI
1170 table underlying Appendix C) are available in the online supplementary
1171 material accompanying this submission.

1172 *Data availability*

1173 Data in support of the findings of this study is available from the corre-
1174 sponding author upon reasonable request, subject to the World Bank Enter-
1175 prise Surveys' own terms of use for the underlying firm-level microdata (enter-
1176 prisesurveys.org, free registration required), including the standardized cross-
1177 country database and Productivity Database underlying the global sample.
1178 The country-level SBFN policy-status (including the SBFN data portal's
1179 member roster underlying the global sample) and Worldwide Governance In-
1180 dicators datasets constructed for this paper, the country-year World Develop-
1181 ment Indicators panel and Callaway-Sant'Anna event-study code underlying
1182 Section 6 (all freely redistributable, no registration required), together with
1183 the full analysis and replication code, are available in a public GitHub repos-
1184 itory; the repository link is withheld from this anonymized manuscript, and
1185 repository access is restricted for the duration of double-anonymized review,
1186 to avoid identifying the authors to reviewers. The link will be provided to

1187 the editor upon submission and made publicly accessible again to readers
1188 upon acceptance.

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